

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Loan Eligibility Prediction	Predicts whether a loan application will be approved or rejected using applicant financial and demographic data.	Core Logic / ML Model	Done	1
2	Applicant Data Input Form	Web form to collect applicant details (income, credit history, employment, etc.) for prediction.	Presentation (Flask templates)	Done	0.5
3	Data Preprocessing & Encoding	Cleans missing values and encodes categorical fields before feeding data to the model.	Training pipeline (Notebook)	Done	0.5
4	Feature Scaling	Standardizes numerical input features using a saved StandardScaler before prediction.	Core Logic (scale1.pkl)	Done	0.5
5	Random Forest Model Training	Trains and compares ML models, selecting Random Forest for final deployment.	Training pipeline (Notebook)	Done	1
6	Prediction Result Display	Displays the loan approval/rejection outcome to the user after submission.	Presentation (submit.html)	Done	0.5
7	Model Persistence	Saves the trained model and scaler as .pkl files for reuse without retraining.	Core Logic	Done	0.5
8	Input Validation & Error Handling	Validates and gracefully handles invalid or missing applicant input.	Core Logic	Pending	0

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, prediction input form, result display page
2	Backend / Logic	4	Data preprocessing, feature scaling, Random Forest prediction, model loading
3	Database / Storage	1	CSV dataset and pickled model/scaler files (local file storage)
4	API / Integration	1	Flask routes (/predict, /submit) handling requests and rendering results
5	Security / Authentication	0	Not implemented in current scope